Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

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Grade and Subject Teacher's Signature	

Task Number	3
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Requirements, Architecture and Technology Setup: Design the minimum application architecture for the Loan Application Tracking Portal. The MVP should cover data/event entry, searchable dashboard, summary indicators, status drill-down and alert/exception view. Select the programming stack, Maven/Gradle/Ant build tool, database if required, Tomcat/Nginx deployment target and local development setup.
Deliverable:	

1. SRS Summary (Software Requirements Specification)

1.1 Functional Requirements (FR)

Req ID	Module / Feature	Specification
FR-01	**Data & Event Entry**	The system must allow loan officers to create a loan application with Applicant Name, Email, Phone, Loan Amount, and Loan Type (*Personal, Home, Auto, Education*). Status defaults to SUBMITTED.
FR-02	**Searchable Dashboard**	The system must provide a unified dashboard listing all loan records with real-time text-based search by Application ID or Applicant Name.
FR-03	**Summary Indicators**	The dashboard must display real-time aggregate metric counters: *Total Applications*, *In Review*, *Approved*, and *Rejected*.
FR-04	**Status Drill-Down**	Clicking any loan record must display a detailed modal or page containing applicant information and an audit timeline of all state transitions with timestamps.
FR-05	**Alert / Exception View**	The system must provide a dedicated view highlighting loans with status `REJECTED` or flagged for exceptions (e.g., *Incomplete KYC*, *Low Credit Score*).
FR-06	**Health Check Endpoint**	The application must expose an automated health check endpoint (`/actuator/health` or `/health`) returning HTTP `200 OK` for DevOps monitoring and smoke testing.

1.2 Non-Functional Requirements (NFR)

Req ID	Category	Specification

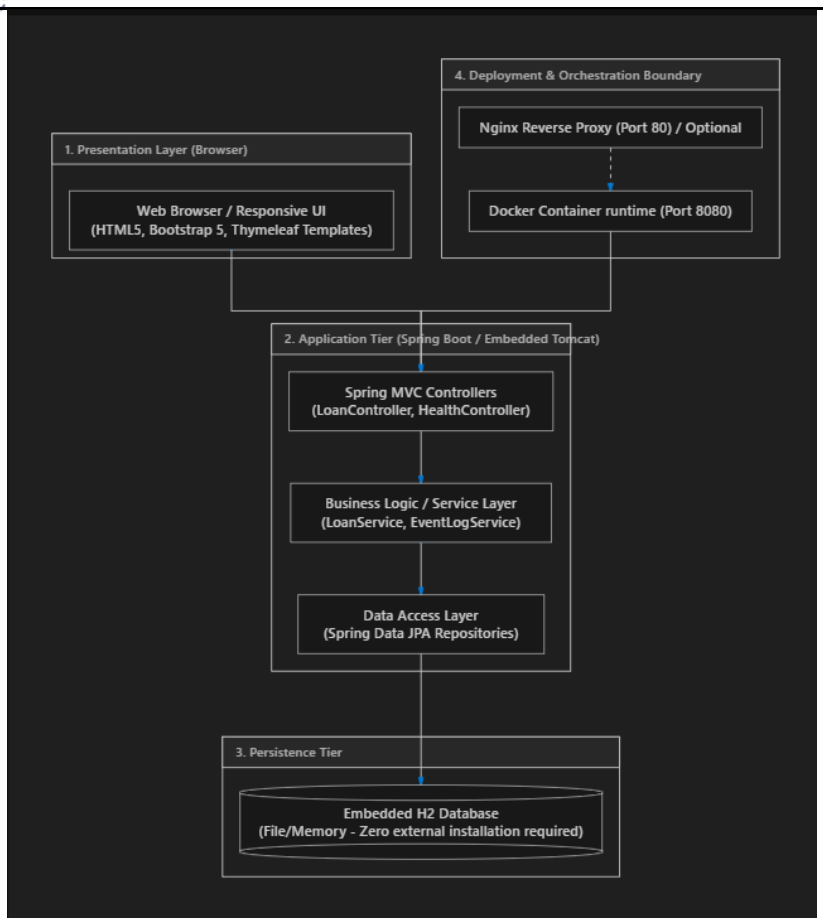
NFR-01	**Performance**	Page load time and search response time must be under **500 milliseconds** for standard workloads.
NFR-02	**Portability**	The application must be packaged as an executable artifact (JAR/WAR) that runs identically on local workstations, inside Docker containers, and on remote Linux servers.
NFR-03	**Security**	PII data must be handled through strict validation; no hardcoded credentials in source control; input sanitization against XSS and SQL injection.
NFR-04	**Testability**	Every UI element (inputs, buttons, alerts, status cards) must have unique, deterministic HTML `id` attributes to enable automated Selenium WebDriver locators.
NFR-05	**Reliability & Audit**	Application state transitions must be immutable event logs stored with timestamps.

2. Use-Case Diagram



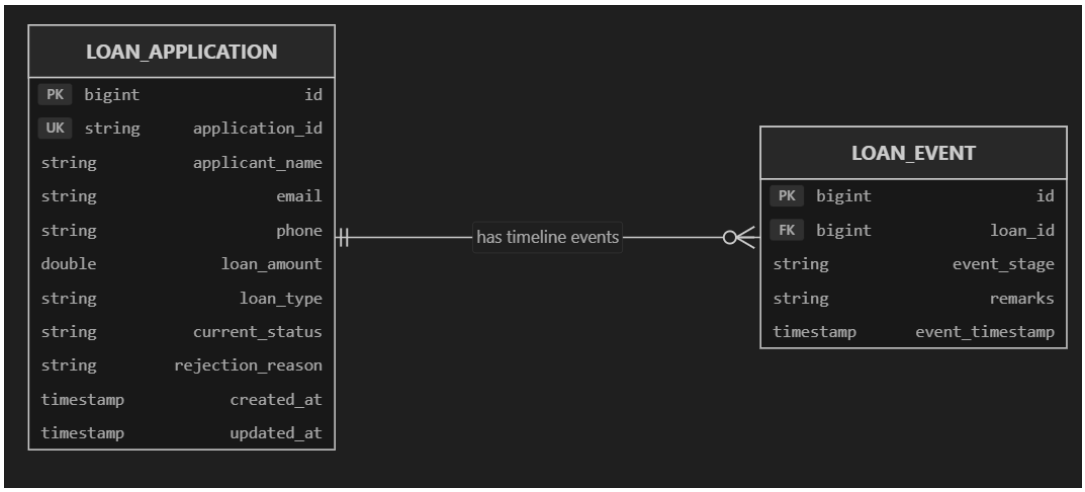
3. Minimum Application Architecture

The application adopts a clean, layered 3-Tier MVC architecture designed for testability, zero-friction local execution, and effortless containerization:



4. Data Model and API / Web Route List

4.1 Data Model (Entity-Relationship)



4.2 Web Routes & REST Endpoints

HTTP Method	Route / Endpoint	Description	Consumes / Produces

GET	/ or /dashboard	Render main dashboard with summary indicators & search table	HTML
GET	/apply	Render new application entry form	HTML
POST	/apply	Create new application & log initial event SUBMITTED	Form Data →|*rightarrow*→ Redirect
GET	/loans/{id}	Status drill-down: view application details and timeline	HTML / JSON
POST	/loans/{id}/status	Update loan stage (e.g. advance to VERIFIED, APPROVED, or REJECTED)	Form Data →|*rightarrow*→ Redirect
GET	/alerts	Render alert/exception view for rejected or flagged applications	HTML
GET	/actuator/health	Automated health check endpoint returning { "status": "UP" }	JSON

5. Technology Stack Selection & Rationale

Layer / Role	Selected Technology	Version	Rationale
Programming Language	**Java**	**Java 17 (LTS)**	Enterprise industry standard, long-term support, seamless Jenkins and Maven integration.
Framework	**Spring Boot**	**3.2.x**	Eliminates boilerplate XML configuration, provides embedded Tomcat server out-of-the-box, Spring Data JPA, and actuator health metrics.
Build Tool	**Apache Maven**	**3.8+**	Standard lifecycle (compile, test, package), dependency resolution, integrates directly with Jenkins CI and Selenium.
UI Templating	**Thymeleaf + Bootstrap 5**	Latest	Server-side rendered UI with distinct DOM IDs (id="applicantName", id="searchBox") crucial for reliable Selenium automation.
Database	**H2 Database**	Embedded	Zero external installation; runs in-memory or as a local file; ideal for CI test execution and containerized demos.
Deployment Target	**Embedded Tomcat / Docker**	Tomcat 10 / Docker 24+	Built into Spring Boot; runs locally on port 8080 without manual server installations; easily wrapped in Docker or reverse-proxied by Nginx.
Automated Testing	**JUnit 5 + Selenium WebDriver**	5.x / 4.x	Standard UI & regression testing with Chrome/Firefox headless support in CI.

6. Local Development Setup Guide

```
Microsoft Windows [Version 10.0.26300.9550]
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C:\Users\vilas>java -version
java version "21.0.12" 2026-07-21 LTS
Java(TM) SE Runtime Environment (build 21.0.12+7-LTS-205)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.12+7-LTS-205, mixed mode, sharing)

C:\Users\vilas>mvn -version
Apache Maven 3.9.16 (2bdd9fddda4b155ebf8000e807eb73fd829a51d5)
Maven home: C:\Dev Tools\apache-maven-3.9.16
Java version: 21.0.12, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21.0.12
Default locale: en_IN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"

C:\Users\vilas>git --version
git version 2.55.0.windows.2
```